

$$6 \quad (a) \quad \lambda = \frac{3 \times 10^8}{75 \times 10^9} (= 0.004 \text{ m})$$

$$\theta_R = 1.22 \frac{\lambda}{b} = (1.22) \frac{0.004}{5.0 \times 10^{-2}} = 0.0976$$

$$\text{max distance} = \frac{7.0}{0.0976} = 71.7 \text{ m}$$

$$(b) \quad (i) \quad 1. \quad I_A = \frac{60}{4\pi(24)^2} = 8.29 \times 10^{-3} \text{ W m}^{-2}$$

$$2. \quad \text{distance BP} = \sqrt{24^2 + 7^2} = 25 \text{ m}$$

$$I_B = \frac{60}{4\pi(25)^2} = 7.64 \times 10^{-3} \text{ W m}^{-2}$$

$$(ii) \quad \frac{I_B}{I_A} = \left(\frac{A_B}{A_A} \right)^2$$

$$\frac{A_B}{A_A} = \sqrt{\frac{7.64 \times 10^{-3}}{8.29 \times 10^{-3}}}$$

$$\frac{A_A}{A_B} = 0.96$$

$$(iii) \quad \text{path difference} = 25 - 24 \quad (= 1.0 \text{ m})$$

$$\Delta\phi = \frac{1.0}{0.40}(2\pi) = 5\pi$$

$$(iv) \quad \left(\begin{array}{l} I \propto A^2 \Rightarrow I = kA^2 \Rightarrow A = \frac{1}{k} \sqrt{I} \\ A_A = \frac{1}{k} \sqrt{I_A}; A_B = \frac{1}{k} \sqrt{I_B} \end{array} \right)$$

$$A_{\text{resultant}} = \frac{1}{k} \left(\sqrt{8.29 \times 10^{-3}} - \sqrt{7.64 \times 10^{-3}} \right)$$

$$= \frac{1}{k} (3.64 \times 10^{-3})$$

$$I_{\text{resultant}} = k (A_{\text{resultant}})^2$$

$$= (3.64 \times 10^{-3})^2$$

$$I_{\text{resultant}} = 1.33 \times 10^{-5} \text{ W m}^{-2}$$

- (c) If the microwaves are not coherent, the phase difference at position P will be changing.

Thus the intensity of the resultant microwave will not be constant.